

Build Your Own AI Voice + Text Assistant (React + FastAPI)

This guide explains step-by-step how to build a simple AI Voice + Text Assistant using:

- React.js Frontend
- FastAPI Backend
- OpenAI GPT API
- Whisper Speech-to-Text
- gTTS Text-to-Speech

The final application will support:

1. Text Chat
2. Voice Input
3. AI-generated responses
4. AI voice replies

Step 1 — Create Project Structure

Create the main project folder:

```
ai-voice-agent/  
├──  
│   ├── frontend/  
│   ├── backend/  
│   │   ├── main.py  
│   │   ├── requirements.txt  
│   │   ├── .env  
│   └── audio/
```

Step 2 — Setup React Frontend

Open terminal and run:

```
mkdir ai-voice-agent  
cd ai-voice-agent  
npm create vite@latest frontend
```

Choose:

- React
- JavaScript

Step 3 — Install Frontend Dependencies

Run:

```
cd frontend  
npm install axios
```

Optional:

```
npm install react-icons
```

Step 4 — Setup Backend

Create backend folder:

```
mkdir backend  
cd backend  
Create virtual environment:
```

```
Windows:
python -m venv venv
venv\Scripts\activate
```

Step 5 — Install Backend Packages

Install required Python packages:

```
pip install fastapi uvicorn openai python-multipart gtts python-dotenv
```

Optional:

```
pip install pydub ffmpeg-python
```

Step 6 — Create requirements.txt

Run:

```
pip freeze > requirements.txt
```

Step 7 — Get OpenAI API Key

Visit:

<https://platform.openai.com>

Steps:

1. Create account
2. Generate API key
3. Copy API key

Step 8 — Create .env File

Inside backend folder create:

```
OPENAI_API_KEY=your_api_key_here
```

Step 9 — Create Backend Server

Create backend/main.py file.

Backend responsibilities:

- Receive user messages
- Send text to GPT
- Convert speech to text
- Convert AI response to voice
- Return results to frontend

Step 10 — Run Backend

Run backend server:

```
uvicorn main:app --reload
```

Backend URL:

```
http://127.0.0.1:8000
```

Step 11 — Create React Chat UI

Frontend features:

- Input box
- Send button
- Voice recording button
- Display AI response
- Play AI audio reply

Step 12 — Run Frontend

Run:

```
npm run dev
```

Step 13 — Voice Processing Flow

Voice Workflow:

```
graph TD
    A[User Voice] --> B[Whisper API]
    B --> C[Text]
    C --> D[GPT Model]
    D --> E[AI Response]
    E --> F[gTTS]
    F --> G[Audio Reply]
```

Step 14 — Add Better UI

Improve UI using:

- Tailwind CSS
- Chat bubbles
- Mobile responsiveness
- Mic animation
- Dark mode

Step 15 — Advanced Features

Future upgrades:

- Chat history
- MongoDB storage
- AI memory
- Streaming responses
- Realtime voice conversations
- PDF chatbot
- Custom AI personalities

Step 16 — Deploy Project

Frontend:

Deploy on Vercel

Backend:

Deploy on Render

Estimated Cost

Using GPT-4.1-mini and Whisper:
Approximate monthly cost:
■300-■1500 depending on usage

Recommended Learning Path

Phase 1:
Build text chatbot

Phase 2:
Add voice input

Phase 3:
Add voice output

Phase 4:
Add AI memory and personalities

Congratulations! After completing this project, you will have: • Your own mini ChatGPT • Voice AI assistant • Full-stack AI application • Portfolio-ready AI project • Real-world experience with AI APIs